



UNDERSTANDING TRANSFORMERS

THE ARCHITECTURE POWERING MODERN AI





HOW TRANSFORMERS CHANGED AI

Since their introduction in 2017, Transformers have become the foundation of modern AI:

- **LLMs like GPT-4 & Claude** → Powering human-like conversation and reasoning.
- **Generative AI** → Driving models like DALL·E and Stable Diffusion for text-to-image creation.
- **New Techniques** → Advances like retrieval-augmented generation (RAG) and fine-tuning improve efficiency and accuracy.

From chatbots to creative AI, Transformers are shaping the future of AI.

Let's dive into how they work.





KEY COMPONENTS

1. Input Embeddings:

Words are converted into numerical vectors so the model can process them. These embeddings capture meaning and context.

2. Positional Encoding:

Transformers don't process words one by one, so they need a way to understand word order. Positional encodings add this information to embeddings.





SELF-ATTENTION & MULTI-HEAD ATTENTION

Self-Attention helps the model determine how important each word is relative to others in a sentence.

For example: In "She unlocked her phone because she forgot the password", "*password*" is more related to "*forgot*" than "unlocked." Self-attention helps the model figure that out.

Multi-Head Attention:

- Instead of focusing on just one relationship, the model runs multiple attention mechanisms in parallel.
- Each "head" captures different aspects of the text (e.g., grammar, meaning, or relationships between words).





FEEDFORWARD LAYERS & RESIDUAL CONNECTIONS

Once self-attention has refined the word representations, they pass through a fully connected feedforward network.

Why? It allows the model to learn more complex patterns.

Residual Connections & Layer Normalization

- Prevent information loss.
- Help with stable and efficient training.





ENCODER-DECODER STRUCTURE

- **Encoder:** Processes the input sentence and generates a rich representation of its meaning.
- **Decoder:** This representation is used to generate output (e.g., a translated sentence).

💡 In machine translation, the encoder processes an English sentence, and the decoder generates the same meaning in French.





WHY TRANSFORMERS ARE SO POWERFUL?

- **Parallel Processing** → Faster training than RNNs.
- **Better Context Understanding** → Thanks to self-attention.
- **Handles Long Sequences** → No vanishing gradient problem like in RNNs.

These advantages make Transformers the go-to architecture for modern AI applications.





BEYOND NLP

🚀 Transformers aren't just for text anymore.

They're now used in:

📷 **Computer Vision** (Vision Transformers - ViTs)

🎵 **Audio Processing** (Speech recognition & music generation)

🧬 **Biology & Drug Discovery** (Protein structure predictions)

